2220 - Entrepreneurial Finance and Venture Capital

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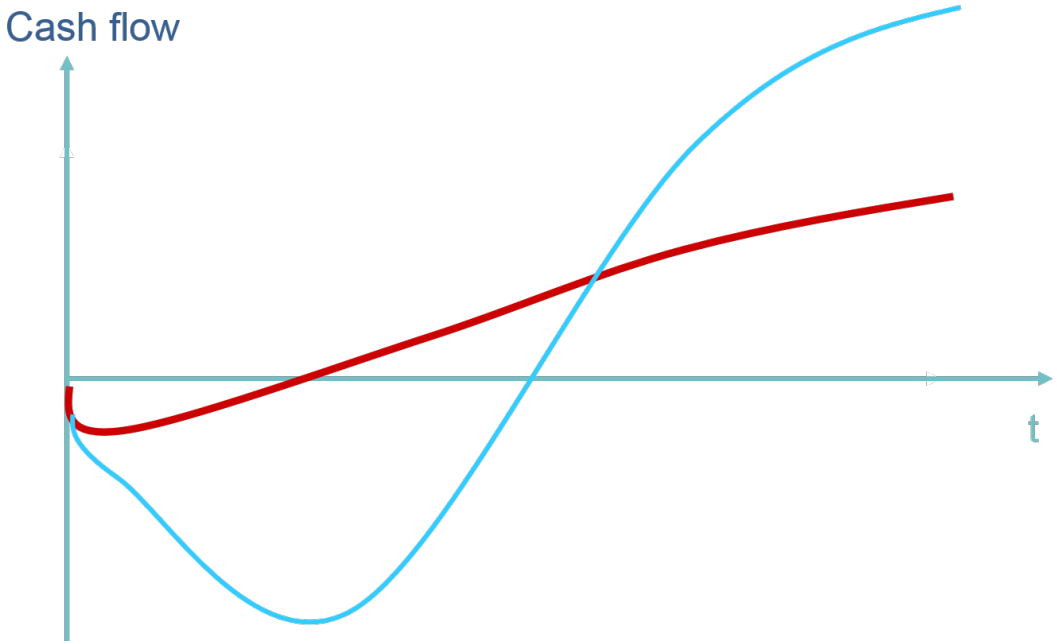
Lecture #3

Business models and cash flow curves



- Different business models and growth trajectories generate different cash flow patterns
- These differences are particularly important in entrepreneurial finance
  - They lead to very different funding needs
  - External finance is expensive for startups

A three-stage, three-parameter model

- Introduce a model to develop understanding of how business models, growth and cash flows are linked
  - Simple, helps build intuition
  - Only requires financial statements data

1. Cash flows are a function of two key business model parameters – profitability and asset intensity – and sales growth
2. Life cycle is split intro three stages: startup, growth and steady-state
3. Profitability, asset intensity and growth are assumed constant within each stage

Input parameters

$$\text{Profitability: } p_t = \frac{\text{EBIAT in year t}}{\text{Sales in year t}}$$

$$\text{Asset intensity: } a_t = \frac{\text{NOA in year t}}{\text{Sales in year t}}$$

$$\text{Growth: } g_t = \frac{\text{Sales in year t}}{\text{Sales in year t-1}} - 1$$

Free Cash Flow formula

- Using these parameters and letting S denote sales we can write

$$\begin{aligned} FCF_t &= EBIAT_t - \Delta NOA_t \\ &= p_t S_t - (a_t S_t - a_{t-1} S_{t-1}) \end{aligned}$$

- Noting that $S_{t-1} = S_t / (1 + g_t)$ we can rewrite this as

$$FCF_t / S_t = p_t - a_t \frac{g_t}{1 + g_t} - \frac{a_t - a_{t-1}}{1 + g_t}$$

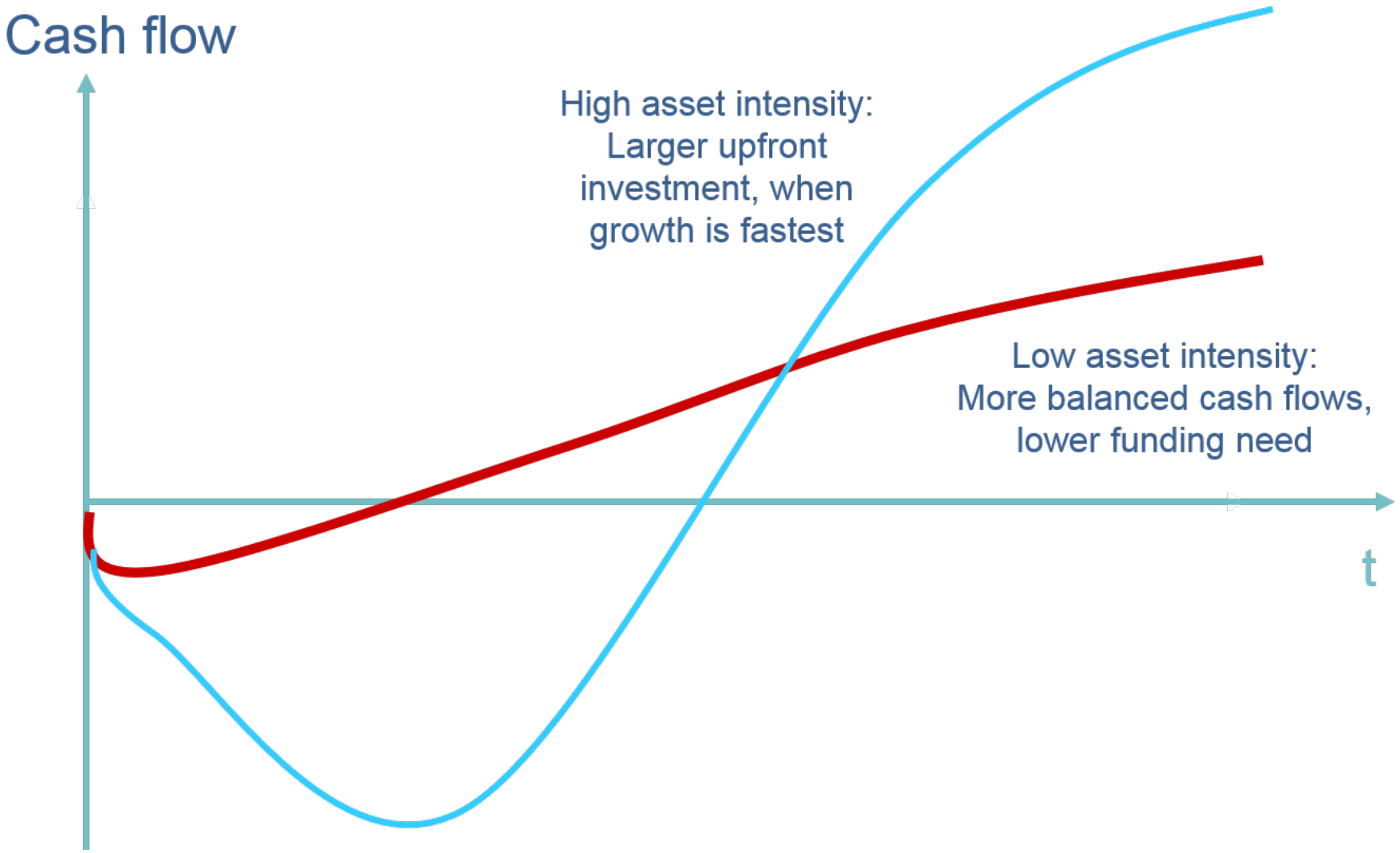
- If $a_t = a_{t-1}$ then we can simplify further

$$FCF_t / S_t = p_t - a_t \frac{g_t}{1 + g_t}$$

Why is this distinction useful?

$$FCF_t / S_t = p_t - a_t \frac{g_t}{1 + g_t}$$

- Profitability p reflects cash flows that scale with the *level* of sales
- Asset intensity a reflects cash flows that scale with sales *growth*
- p and a have very different implications for the shape of cash curves in high-growth ventures


Cash flow
High asset intensity:
Larger upfront
investment, when
growth is fastest
Low asset intensity:
More balanced cash flows,
lower funding need
t

Goal for today

- Build intuition
  - What business model choices determine p and a?
  - What do these parameters look like for different industries?
  - How do p, a and g interact to determine cash flow curves, funding needs and valuation?

- Note this is a simplified model, but it can easily be extended
  - Annual/monthly values
  - Fixed costs or CAPEX